

RELATIONAL PROBES CAN BE RIGHT FOR THE WRONG REASONS: IDENTIFYING EVIDENCE IN IMAGE GENERATORS

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ABSTRACT

A successful representation probe shows that a target variable is predictable from model activations, but not which evidence supports that prediction. This identification problem is especially severe for relational variables, whose labels can correlate with entity identity, local appearance, global structure, or information propagated from elsewhere in the input. We study graph-role probes in Qwen-Image and Stable Diffusion 3.5 using matched scientific diagrams in which two visually identical edges receive different roles from the surrounding graph. Across successive controlled designs, apparent graph-role signals arise from endpoint identity, global chain length, and a biased estimator reference. Standard construction-validity and label-shuffle controls do not identify the first two sources; the shuffle control detects only the displaced estimator. After these alternatives are controlled, a strong Qwen-Image readout remains in one pre-registered subgroup. Although the readout is local, selectively blocking its access to a remote role-defining arrowhead reduces the role estimate ΔS from +0.730 to +0.005 bits, while a size-matched control cut preserves +0.636. A precommitted graded intervention shows the signal declining sharply as access is progressively removed, with the construction-validity control intact throughout. These results motivate a relational-probing battery combining nuisance-controlled stimuli, empirical estimator nulls and graded access interventions, and support claims only about decodability under the tested readouts — not graph understanding or use during generation.

1 INTRODUCTION

Representation probes establish whether a variable is decodable from an activation, but not which evidence makes it decodable, nor whether that information is used behaviorally (Belinkov, 2022; Hewitt & Liang, 2019; Ravichander et al., 2021). The gap opens whenever the label correlates with something the experimenter did not intend to measure.

Relational variables make it unusually wide. A graph role is a property of surrounding structure, so the label is defined somewhere other than where it is read, and what actually carries the signal can be the identity of the entities an edge connects, its position and local geometry, a global property such as chain length that covaries with role, or visual evidence propagated from elsewhere in the input. These are not arbitrary bugs: each is a different *kind* of mechanism, and each calls for a different instrument.

mechanism	kind
endpoint identity	local stimulus correlate
global chain length	global stimulus correlate
displaced estimator null	measurement artifact
remote arrowhead	model-mediated visual evidence

Only three are sources of predictive *evidence*; the displaced null is an artifact of how the estimate was computed. The distinction matters because it determines which instrument removes each one.

Standard practice supplies two controls: a construction-validity gate, asking whether the same features, folds and classifier family recover a contrast the design is known to contain, and a label-shuffle

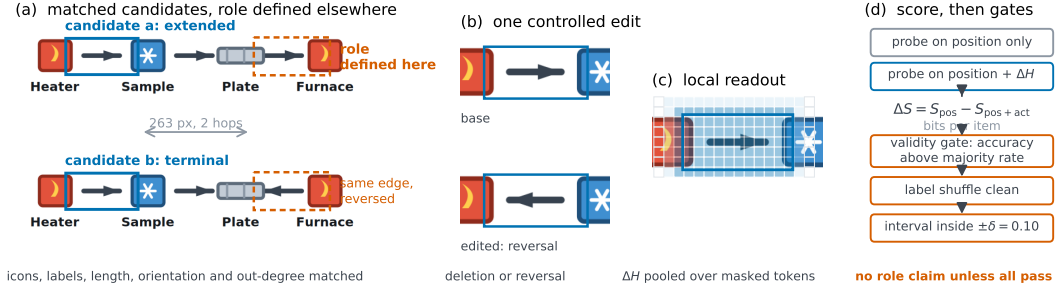


Figure 1: **The controlled setup.** (a) The two candidate edges are drawn identically — same icons, labels, length, orientation and target out-degree — so role is fixed only by structure elsewhere, 263 pixels and two graph hops away; a third control lane, omitted for space, carries the validity contrast. (b) Base and edited diagrams are rendered independently from their graphs, differing in exactly one candidate edge. (c) The readout pools the activation difference over the edited-edge tokens and a one-cell ring, never overlapping the role-defining arrow. (d) ΔS is the gain over a position-only probe, read only after all three gates pass.

control, asking whether the estimate collapses under permuted labels. Both are necessary and neither is designed for source identification — the first tests that the readout works, the second that the estimator responds to randomized labels, and a readout predicting the nominal role label from a nuisance correlate satisfies both. They are insufficient here rather than defective, and this paper is about what has to be added.

A testbed where the label is defined outside the readout. Scientific diagrams place the label-defining evidence and the measurement in separate regions. Two candidate arrows are matched in appearance, endpoint identity, relation type, orientation and local geometry, and take different graph roles only through a third arrow roughly 263 pixels away, outside the readout mask (Figure 1). We edit one candidate edge, re-render, and probe the activation change at the candidate against a position-only baseline, holding out complete layouts in cross-validation so a probe cannot succeed by memorizing where things sit.

Three mechanisms, and one signal that survives. Successive designs expose three of them. End-point identity predicts the nominal role even when both standard controls pass. After identity is removed, a higher-capacity decoder instead recovers global chain length — a variable defined outside the matched radius, where no locally defined control can see it. A separate per-item estimator produces an apparent effect because its empirical no-information point is displaced from zero, the one case the shuffle control is built to detect. Redesigning the stimuli and calibrating the estimator remove all three, after which one strong Qwen-Image signal remains, and the question is what supports it.

From rendering evidence to an intervention on access. Holding the graph fixed and moving the role-defining arrow farther away leaves the estimate intact; changing how the same arrow is rendered nearly eliminates it. That implicates remote visual evidence but cannot settle the question, because two differently rendered stimuli are not guaranteed comparable in any other respect. So we intervene on the model. Blocking candidate-region queries from attending to the remote arrowhead at every transformer block reduces the estimate from +0.730 to +0.005 bits, while a size-matched cut over a control region preserves +0.636 and construction validity remains recoverable; the same cut is inert where there is no role signal to remove, and it removes SD3.5’s much smaller estimate. A binary ablation tells us that access matters but not how much, so we then grade the cut, removing a fraction of the region’s tokens chosen independently of attention rank and reading each dose against its own condition’s unmasked baseline. That measures dependence on the *quantity* of access without assuming separately rendered stimuli are otherwise comparable, and without treating attention weight as an explanation — and because the mask applies at every block, it identifies an access dependency, not a minimal circuit.

The battery, stated compactly. Taken together the instruments form a staged procedure, each stage being what remains when the previous one is exhausted:

1. Break identifiable nuisance correlations by construction.
2. Measure the estimator’s empirical null rather than assuming it is zero.
3. Hold the intended structure fixed while varying candidate evidence.
4. Intervene directly on access to the implicated evidence.
5. Grade that intervention relative to each condition’s own baseline.

Our contributions are:

- **Identification failure modes.** We document four mechanisms producing an apparent relational probe signal — two stimulus correlates, one measurement artifact, and one genuine model-mediated dependence — and show why no single standard control separates them.
- **A diagnostic battery.** We formulate the five-stage battery above, so that the procedure rather than this model’s behavior is the transferable object.
- **A graded-access test, and a bounded finding.** We introduce a within-condition graded-access test and use it to show that Qwen-Image’s strong local readout in the identified subgroup depends sharply on access to a remote role-defining arrowhead. The result concerns decodability in that subgroup; it does not establish a general graph representation or its use during generation.

2 RELATED WORK

Probing, selectivity, and causal interpretation. Recoverable structure, as in Othello-GPT (Li et al., 2023), motivates probing for relational representations (Greff et al., 2020). Probe success nevertheless establishes decodability (Belinkov, 2022), which can persist without task necessity or behavioral importance (Ravichander et al., 2021; Elazar et al., 2021). Control tasks test whether a probe is selective (Hewitt & Liang, 2019), and probe capacity itself affects information estimates (Pimentel et al., 2020; Voita & Titov, 2020). Our gap is one step further in: not whether a probe is selective for its label, but which evidence in the input supports the prediction it makes.

Shortcut learning and control tasks. Shortcut learning is the broader precedent: models succeed using rules that fail under harder conditions (Geirhos et al., 2020), classifiers exploit artifacts such as source tags (Lapuschkin et al., 2019), labels are partly predictable from hypotheses alone (Gururangan et al., 2018), and diagnostic datasets are built to expose such heuristics (McCoy et al., 2019). Interpretability methods are not immune: saliency maps can be insensitive to the model they explain (Adebayo et al., 2018) and conclusions about explanation methods can depend on evaluation artifacts (Yona & Greenfeld, 2021). The shortcuts we document are of this kind, but they are nuisance correlations of a *relational* label, which is what makes them invisible to a control defined on the locally matched region.

Causal interventions, and attention as evidence. Inside the generator, causal mediation, layer-wise decoding and cross-attention interventions localize and repair object, style, colour and attribute effects (Basu et al., 2024; Toker et al., 2024; Hertz et al., 2023; Chefer et al., 2023). Most directly related, Wang et al. (2026) train diffusion transformers on two-object prompts with a spatial relation and recover an explicit relation circuit; their relation is spatial and realized through object placement, whereas we use repeated, visually matched instances of the same relation and condition explicitly on position, so entity identity cannot determine which instance occupies which role. Our intervention also serves a different purpose: it identifies the evidence a probe depends on, not a minimal circuit. A separate literature asks whether attention weights themselves explain (Jain & Wallace, 2019; Wiegrefe & Pinter, 2019), tests that by erasure (Serrano & Smith, 2019), or refines attribution by value norm (Kobayashi et al., 2020) and layer-wise propagation (Abnar & Zuidema, 2020). We take no position there: our cuts restrict *access* at fractions chosen independently of attention rank, so they measure dependence on a source rather than rank attention weights. Masking does renormalize attention and change downstream activations, which is part of the intervention.

Relational structure in image generators. Relational and binding information is lost by CLIP’s text encoder as captions become more compositional, and binding failures have been traced to it (Kamath et al., 2023b; Zarei et al., 2024; Wu et al., 2024). Relational structure remains a persistent

failure mode for these models more broadly (Thrush et al., 2022; Kamath et al., 2023a; Yuksekogonul et al., 2022; Gokhale et al., 2022; Huang et al., 2023; Lin et al., 2026; Luo et al., 2026; Zhang et al., 2026; Zala et al., 2024; Wei et al., 2025), but those works establish failures in outputs; our use of diagrams is methodological.

3 METHOD

The procedure is a battery applied in order: stimuli that break each nuisance correlation, an estimator checked against its own null, then holding the graph fixed while varying a candidate visual source and intervening on access to it. The outcome measured throughout is predictive improvement from activations beyond a position-only baseline.

Models and extraction. We study **Qwen-Image**, a 20B-parameter MMDiT with 60 blocks (Wu et al., 2025), and **Stable Diffusion 3.5 Large**, an 8B-parameter MMDiT with 38 (Esser et al., 2024), both on 1024×1024 images. Renoising strength s selects the denoising timestep; the confirmatory extraction used $s = 0.2$ for Qwen-Image and $s = 0.4$ for SD3.5, and the model-by-strength grid was completed afterwards. Depth grids and setting provenance are in Appendix B.

3.1 IS THE LABEL PREDICTABLE FROM ANYTHING LOCAL?

Graph role is the topological function of an edge in its surrounding graph. Our corpus contains 112 base items across seven templates, each with two candidate edges matched in appearance, end-point identity, relation type, orientation, and local geometry. Their roles depend on graph structure outside the candidate region. Global chain structure is crossed with role, and position is explicitly modeled. A visually distinct control edge supplies the construction-validity contrast. Items violating the matching constraints are rejected before extraction; Figure 1 and Appendix C specify the construction.

For each graph G , we render a base image and an edited graph G' differing in exactly one candidate edge by deletion or reversal. The two forward passes share the prompt, noise realization, model, timestep, and extraction block. At block ℓ we compute

$$\Delta H^{(\ell)} = H_{\text{edit}}^{(\ell)} - H_{\text{base}}^{(\ell)}.$$

The primary readout pools tokens overlapping the edited-edge bounding box and a one-token dilation ring. It never directly includes the role-defining remote arrowhead.

3.2 IS THE NULL AT ZERO, AND IS THERE INFORMATION BEYOND POSITION?

We fit a position-only probe and a position-plus-activation probe on identical rows and cross-validation folds, reporting

$$\Delta S = S(\text{label} \mid \text{position}) - S(\text{label} \mid \text{position}, \text{activations})$$

in bits per item. Positional features include box center, size, aspect ratio, orientation, quantized cell, and mask fraction. Positive ΔS measures predictive improvement; it does not identify its source. Folds hold out complete layouts, and 90% confidence intervals use cluster bootstrap over base items. Primary role probes use the pooled estimator; Appendix G.1 audits the earlier per-item estimator.

Three controls govern interpretation. The **construction-validity gate** requires the same features, folds and classifier family to recover the candidate-versus-control contrast above both the majority baseline and the information criterion; a failed gate makes a role estimate unreadable. The **label-shuffle control** repeats the procedure with permuted labels, and intervals excluding zero are flagged. A clean estimate is **negligible** only if its interval lies within $[-\delta, +\delta]$, $\delta = 0.10$ bits; wider intervals are inconclusive (Lakens, 2017; Lakens et al., 2018).

3.3 DOES THE SIGNAL DEPEND ON A REMOTE REGION, AND HOW MUCH OF IT?

The role-defining arrow lies $n - 2$ graph hops from the candidate. Holding the graph, hop count, identities, directions and rendered candidate edge fixed, ten arms vary that arrow’s distance over

238–443 pixels, its arrowhead coverage, shaft length and icon crowding, with displaced-patch and redrawn-on-top controls (Appendix D).

To test the implicated dependency, we mask candidate-region query connections to remote arrowhead keys at every transformer block, under three conditions: `none` (empty mask, a verified pass-through), `head` (target arrowhead), and `ctrl` (a comparable number of keys over the control lane’s arrowhead). Other query rows are not directly masked, but renormalization and downstream changes remain part of the intervention. These runs use three of six layouts; Appendix G.4 audits the position-baseline shift.

We then grade the target cut using $f \in \{0, 0.25, 0.50, 0.75, 1\}$, the fraction of arrowhead tokens removed — uniform subsets, fixed seed per item and layout, chosen independently of attention weights. Endpoints reuse the existing runs; the three interior doses and five predictions were committed before extraction (Amendment 20). Every dose uses the same readout and interpretation controls, and the curve measures sensitivity to access quantity without assigning additive contributions to individual tokens.

Supporting measurements. We also record raw attention enrichment for the pooled queries, and run an MLP, pairwise endpoint pooling, whole-sequence readouts where valid, and 349 AI2D-RST textbook diagrams (Hiippala et al., 2021; Kembhavi et al., 2016). All are descriptive or corroborative (Appendices H and G.5); no claim rests on them, and every claim concerns recoverability under the tested procedures.

4 RESULTS

Unless stated otherwise, estimates are fitted on all six layout permutations of each arm (1008 rows). The masking experiments of Sections 4.4–4.5 use three permutations and are read only through within-condition contrasts; the resulting position-baseline shift is audited in Appendix G.4, and Table 7 converts between the two estimators cell by cell, showing the difference is the position marginal rather than a perturbed forward pass.

4.1 A STRONG APPARENT GRAPH-ROLE SIGNAL

The confirmatory grid contains one clear apparent positive. Under arrow reversal at low renoising strength, the strongest Qwen-Image cell over the full 112-item set is block 47, at 0.723 accuracy and $\Delta S = +0.253$ bits, $CI_{90} [+0.217, +0.288]$, with the validity gate passing, a clean label shuffle, and independent recovery by a whole-sequence readout. The effect occupies a late band rather than a single block (Figure 5A): blocks 9–32 stay inside the equivalence region while 35–47 yield $+0.167$ to $+0.382$ bits, gate-passing and shuffle-clean throughout. It is also condition-specific — deletion stays below 2.2×10^{-4} bits across the same blocks, and reversal at $s = 0.4$ stays inside the equivalence region wherever the measurement is readable.

The signal is carried almost entirely by the shorter graphs, and comparisons must hold the block fixed. At block 38 the full 112-item set gives $+0.382$ bits at 0.794 accuracy, the 56-item $n = 4$ subgroup $+0.935$ bits at 0.992, and $n = 5$ only $+0.017$ at 0.562. Every source-identification experiment below runs within that subgroup at block 38 against the $+0.935$ -bit reference; the block 47 figures above are a different measurement, not a weaker version of the same one.

Taken alone this would appear to support a strong graph-role readout. Three earlier designs produced estimates that looked as convincing and were not graph role at all; the corpus above exists because each was killed.

4.2 THREE SIGNALS THAT WERE NOT GRAPH ROLE

The three cases below are in the order they arose, and Figure 2 plots them alongside the signal that survives. Each was removed by a different instrument — two by stimulus redesign, one by measuring the estimator’s own null (Appendix G.1).

Endpoint identity. Before endpoint identity was controlled by construction, a probe separated the two candidate roles at $\Delta S = +0.539$ bits. Restricting the analysis to the one family whose candidate

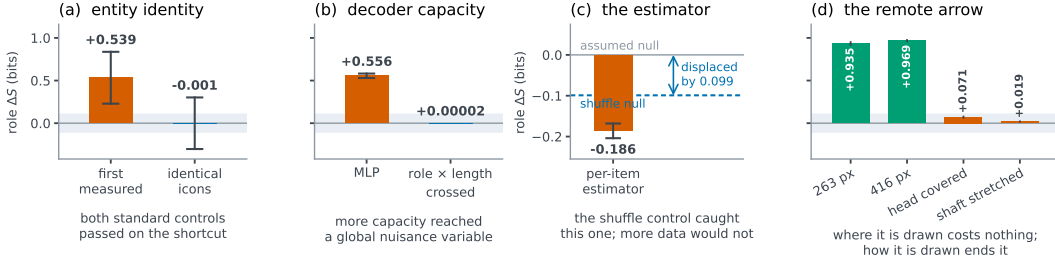


Figure 2: **Four mechanisms producing an apparent graph-role signal, and what separated each one.** Role ΔS in bits; whiskers are 90% cluster-bootstrap intervals and the shaded band is the pre-specified $\pm\delta = 0.10$. **(a)** Endpoint identity, collapsing on the 32-item identical-icon family, whose interval is correspondingly wide; both standard controls pass on the confounded design. **(b)** Global chain length, reached by a higher-capacity decoder and removed by crossing role with lane length. **(c)** The per-item estimator, whose label shuffle sits at -0.099 rather than zero, so the equivalence test has no reference — the one of the three a standard control is built to detect. **(d)** Remote rendering evidence, the source that survives: block 38 over six layouts, where 153 pixels of extra distance cost nothing and two unrelated changes to how the same arrow is drawn end the signal.

edges use identical endpoint icons and labels reduces this to -0.001 bits on 32 items — excluding an effect of the original magnitude without establishing a negligible one. Both standard controls pass cleanly in the confounded design, because both are compatible with a readout that predicts the nominal role label from endpoint identity.

Global chain length, amplified by decoder capacity. With identity removed, a nonlinear probe produced a larger apparent signal than the linear one: an MLP recovered $\Delta S = +0.556$ bits against $+0.006$ for a linear probe on the same rows. Crossing graph role with total lane length by construction removes it — the same MLP at the same block and timestep falls to $+0.00002$ bits. The candidate arrows were locally matched throughout; the exploited variable is global chain length, recoverable at the local edit site but defined outside the matched radius, where no locally defined control can see it.

A displaced estimator, which a standard control did catch. An early confirmatory analysis returned $\Delta S = -0.186$ bits on a cell whose validity gate passed decisively. The estimate was precise and excluded zero, but the label-shuffle distribution was centered near -0.099 , so the equivalence test had no reference. This is bias, not sample size — more data narrows the interval around the displaced centre. Fitting a separate probe within each item produced it; all reported results use the pooled estimator. This is the one case the standard set is built for, and it caught it.

Two criteria also failed: scored on information alone the validity gate is passed at $+0.128$ bits by an image readout classifying at exactly the majority rate, so every gate here also requires accuracy above that rate; and cross-edit transfer failed its own measurement controls and is withdrawn (Appendix G.3).

4.3 THE SIGNAL DEPENDS ON HOW REMOTE EVIDENCE IS RENDERED, NOT WHERE IT IS DRAWN

Holding the graph, hop count, entity identities, edge directions and rendered candidate edge fixed, we change only the remote arrow; Figure 3a reports the ten resulting arms at block 38 and Table 3 the full five-block comparison.

Changing distance does not explain the signal. Placing the role-defining arrow 263, 314, 365 and 416 pixels away while preserving the relevant local rendering gives $+0.935$, $+0.792$, $+0.924$ and $+0.969$ bits (Figure 2d). The farthest condition is among the strongest across the band, and no monotonic attenuation appears over the tested 238–443 pixel range.

Independent re-renderings nearly eliminate the readout. Changing how the same arrow is drawn gives a different pattern. Covering the arrowhead takes the block 38 estimate from $+0.935$ bits to $+0.094$, $+0.081$ and $+0.071$ at 35%, 68% and full coverage, while moving the identical patch

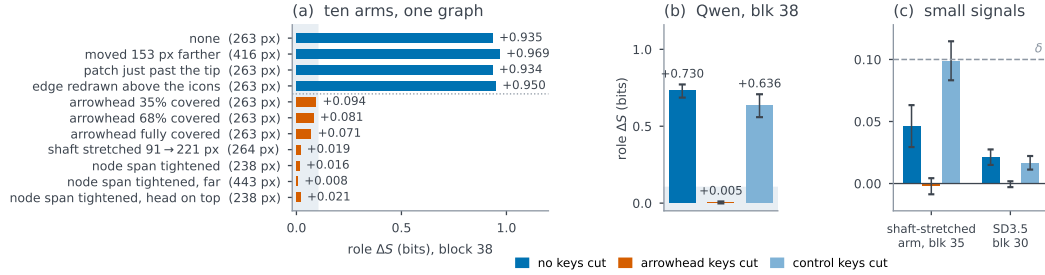


Figure 3: **The source-identification argument.** (a) Ten rendering interventions at block 38, with graph, hop count, entity identities and rendered candidate edge fixed; conditions above the dotted rule vary where the remote arrow is drawn, the rest vary how. (b) Candidate-region queries blocked from attending to the role-defining arrowhead, against an empty mask and a size-matched control cut. (c) The same intervention where the available role signal is small. Shading is the pre-specified $\pm\delta$; panels (b) and (c) use three layout permutations and are read within panel (Appendix G.4).

just past the arrow tip leaves +0.934, so the patch itself does not explain the collapse. Nor is the effect specific to coverage: stretching the shaft from 91 to 221 pixels at about the same distance yields +0.019 bits despite a fully exposed arrowhead, tightening the span between the connected icons yields +0.016, and redrawing the role edge above the icons restores measured exposure to 0.95 yet still leaves +0.021 at the tight span. The arms therefore implicate remote rendering evidence without isolating a single sufficient visual property, and several of them fail the shuffle criterion by -0.003 to -0.009 bits — too small and wrongly signed to explain estimates of +0.07 to +0.09, but enough to limit what an image intervention can carry (Appendix G.2). The causal claim rests on the model intervention next.

4.4 ACCESS TO THE REMOTE ARROWHEAD IS CAUSALLY NECESSARY

The rendering arms implicate the remote arrow but cannot establish how its information reaches the candidate, because two differently rendered stimuli are not guaranteed comparable in any other respect. We therefore intervene on the model, preventing candidate-region queries from attending to tokens covering the role-defining arrowhead at every transformer block, against an empty mask through the same code path and a size-matched cut over the control lane’s arrowhead.

Removing access to the role-defining arrowhead eliminates the apparent signal throughout the late band while the control cut largely preserves it (Table 11, Appendix H, which prints all three conditions at all five blocks): at block 38 the estimate falls from +0.7301 to +0.0046 bits against +0.6357 under the control, and at block 47 from +0.9432 to -0.0048 against +0.9459. Construction-validity accuracy stays between 0.951 and 0.980 across all three conditions, so the cut removes the role information rather than disabling the readout. It is inert where there is no role information to remove: in the shaft-stretched arm it moves the estimate by -0.006 to 0.000 bits, gate still passing, despite strong attention to the arrowhead there.

Two unregistered observations qualify the shaft-stretched control at block 35, both in shuffle-flagged cells; Appendix M reports them and the untested renormalization account, and we do not treat them as confirmatory.

The same intervention removes the small SD3.5 signal. SD3.5 Large has no role estimate outside the equivalence region anywhere in the study, but its deepest readable blocks carry +0.0212 bits at block 30 of 38. The target cut takes that to -0.0005 at chance while the size-matched control leaves +0.0168 — the same access dependency, though not an independent replication.

4.5 A DOSE-RESPONSE CURVE FOR ACCESS TO REMOTE EVIDENCE

A binary ablation shows that access matters but not how much, so we graded the cut: the candidate-edge queries lose a fraction f of the arrowhead’s tokens at every block, $f = 0$ and $f = 1$ being the conditions already reported. Which tokens go is decided uniformly at random per item and layout

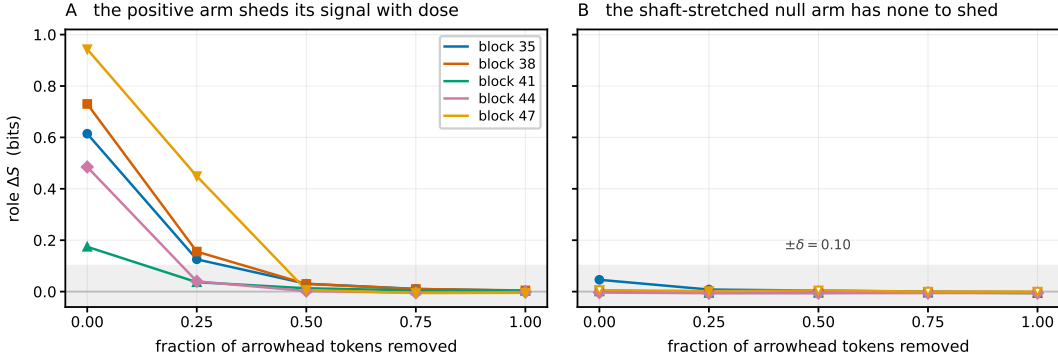


Figure 4: **A dose-response curve for access.** Role ΔS against the fraction of role-arrowhead tokens removed from the candidate-edge queries, one line per band block, three layout permutations. (A) The positive arm sheds its signal steeply, monotonically at blocks 35, 38 and 41 and reversing by under 0.003 bits on the final step at 44 and 47, between cells at chance. (B) The null arm, same axes, has nothing to shed. Open markers are shuffle-flagged; the validity gate passes at all fifty cells.

from a fixed seed, independently of attention, so the attention mass removed is proportional to f only in expectation. The three interior doses, the five predictions and the decision rules were registered before extraction (Amendment 20).

All five predictions held. In the positive arm the estimate falls monotonically at blocks 35, 38 and 41; at block 38 it runs $+0.7301$, $+0.1557$, $+0.0299$, $+0.0101$, $+0.0046$ bits, a total loss of $+0.726$. Blocks 44 and 47 decline the same way and then rise on the final step by 0.0028 and 0.0008 bits, between cells at chance with almost entirely overlapping intervals; we report the reversals rather than call the curve clean. In the null arm every dose lies inside $[-\delta, +\delta]$, with a block-38 span of $+0.008$ bits against $+0.726$, and the validity gate passes in all fifty cells. The decline is steepest at low doses, so no single loss-per-unit applies: a quarter of the tokens costs about four fifths of the signal at blocks 35, 38 and 41, and interpolating to half the loss gives descriptive half-loss doses of 0.156 , 0.158 , 0.155 , 0.137 and 0.240 across blocks 35–47 — summaries of the measured curve, not sampled thresholds.

Every dose is read against its own condition’s $f = 0$ estimate, so the curve never requires two differently drawn stimuli to be otherwise comparable — the assumption the rendering arms cannot supply, and the reason those arms can only implicate a region while this one measures the dependency on it. Because selection is independent of attention weights, this measures sensitivity to access quantity and not whether attention magnitude ranks tokens by causal importance; and ΔS remains predictive improvement under the specified probe, not information used to generate an output.

4.6 WHAT REMAINS, AND THREE READINGS WE WITHDRAW

With the redesigns of Section 4.2 in place, the same features and folds recover construction validity at $+0.915$ bits in Qwen-Image and $+0.908$ in SD3.5 while the role estimates are $+3 \times 10^{-5}$ and $+3 \times 10^{-6}$ bits at chance. A second topological family and additional readouts find no useful role signal under their own controls, though that family’s weaker validity contrast means its null does not establish generality (Appendix N).

Three further readings of our own fell the same way, and they illustrate one principle rather than three lapses: each was consistent with the data available when it was made, and each fell to a measurement that varied the thing it named. A spatial-range account fell to the distance sweep, which finds no decay; an arrowhead-exposure account fell to measuring exposure on the finished images, where unoccluded arrowheads null as hard as the rest; and reading the shaft-stretched arm as the strongest attention routing fell to re-running it at scale. What remains is a dependency on remote rendering evidence with no uniquely sufficient visual property isolated, which we state as a limitation rather than a finding (Appendix O).

5 DISCUSSION

5.1 WHAT THIS CHANGES ABOUT PROBING PRACTICE

The transferable object is the five-stage battery of Section 1, not this model’s behavior, and six things follow for anyone probing a relational variable. **Probe accuracy is not source identification:** our first design reached +0.539 bits with every standard control passing on a readout reading endpoint icons. **Greater decoder capacity can expose stronger nuisance information** rather than a better-encoded target — which is what a nonlinear probe beating a linear one on a global correlate looks like from the inside. **A local readout can depend on remote evidence**, so the region a probe is fitted on does not bound where its information comes from. **Empirical nulls should be measured, not assumed. Interventions should target the hypothesized evidence source**, because rendering two stimuli differently changes more than the property under test. And **a graded intervention beats one-shot deletion**: it separates an all-or-nothing dependency from a continuous one, and its within-condition baseline removes a comparability assumption the rendering arms cannot supply. None of these failure modes has to generalize across models to be worth avoiding.

None of this depends on attention magnitude: enrichment and decodability do not order the rendering arms alike at blocks 35 and 38, and those comparisons are descriptive diagnostics (Appendix L), not a test. Global propagation is expected in an MMDiT, so the informative result is not that remote evidence reaches the candidate but which evidence the readout depends on and how much access it needs.

5.2 BOUNDARIES

The strong signal and the dose–response are established in one pre-registered Qwen-Image subgroup at low renoising strength; SD3.5 shows the same access dependency but its estimate never leaves the equivalence region, so it is not an independent replication. The primary testbed is synthetic and AI2D-RST lacks its matching guarantees. Because the mask applies at every block we identify an access dependency, not a minimal circuit, and the interpolated half-loss doses describe the intervention response rather than a fixed subset carrying that fraction of the information. A negligible controlled estimate bounds recoverability under the tested readouts; it does not show graph structure is absent, and a successful role probe would not have shown use during generation either (Ravichander et al., 2021; Elazar et al., 2021).

Masking uses three of six layouts with a documented baseline shift, shuffle-flagged cells stay marked, and two unregistered masking observations remain exploratory (Appendix M). Nothing here tests whether attention magnitude ranks tokens by causal importance. No single measured visual property explains all rendering effects, and the confinement to low renoising strength is unresolved — of 105 reconstruction failures 97 involve arrows, often an arrowhead dissolving into a bare curve, which suggests but does not establish a loss of the supporting detail as s increases.

6 CONCLUSION

A relational probe can be highly accurate for several non-relational reasons. In one design, two stimulus correlates, a measurement artifact and a genuine dependence on remote visual evidence each produced an apparent graph-role signal, and only the artifact lies inside what the standard controls are built to detect. The dependence that survives every redesign — on how a role-defining arrow outside the readout mask is drawn — is established by a selective cut and quantified by a graded one. Identifying what supports a probe’s prediction therefore requires controlled stimuli, calibrated nulls, and interventions on the model’s access to candidate evidence — not accuracy, validity and label-shuffle tests alone.

REPRODUCIBILITY STATEMENT

We release the stimulus generator and rendered corpus (all ten remote-evidence arms and the three key-masking conditions), the masking wrapper, and the extraction and probing code — position-only baseline, validity gate, label-shuffle control, equivalence test and every readout variant — with

frozen result files for each reported cell and the pre-registered plan with its dated amendments, so every diagnostic can be checked against what was committed before the run. Appendices B and F give protocol and per-cell detail.

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A APPENDIX

This appendix provides the full experimental specification underlying the identification analyses in the main paper. We report the activation-extraction protocol, stimulus construction and validation criteria, the complete spatial-distance intervention, per-cell results, additional probe and control diagnostics, corroboration on naturally occurring textbook diagrams, and the provenance of the confirmatory and diagnostic analyses.

B FULL EXPERIMENTAL PROTOCOL

Both models are run at 1024×1024 resolution with a 64×64 image-token grid. Activations are captured at the transformer-block output. Renoising strength s selects the denoising timestep according to

$$t = \text{ts}[\text{steps} - \min(\text{steps} \cdot s, \text{steps})],$$

so s is a timestep selector rather than a separate model hyperparameter.

The confirmatory grid samples four relative depths, $\{0.15, 0.30, 0.55, 0.80\}$, of each transformer stack. These correspond to blocks $\{9, 18, 32, 47\}$ for Qwen-Image and $\{6, 11, 20, 30\}$ for Stable Diffusion 3.5 Large. Diagnostic sweeps associated with the remote-evidence analyses add relative depths $\{0.60, 0.65, 0.70, 0.75\}$, corresponding to Qwen-Image blocks $\{35, 38, 41, 44\}$ and SD3.5 blocks $\{22, 24, 26, 28\}$.

The controlled corpus contains 112 base items with two rendering seeds per layout. Cross-validation folds hold out complete layouts, with both seeds of a layout assigned to the same fold. Confidence intervals are obtained by cluster bootstrap over base items at the 90% level. All reported graph-role probes use the pooled estimator.

Renoising-strength provenance. The confirmatory extraction paired Qwen-Image with $s = 0.2$ and SD3.5 with $s = 0.4$. This pairing was not selected by a pre-specified rule, and the reconstruction criterion instead favors approximately $s = 0.4$ for both models. We therefore completed the missing cells of the 2×2 model-by-strength design.

After the identified confounds are controlled, every readable cell at either strength, for either model and edit type, lies inside the pre-specified equivalence region. The original model-strength pairing is therefore not load-bearing for the final controlled result.

C CONTROLLED STIMULI AND VALIDATION

The synthetic corpus contains 112 base items across seven templates. Each base item is rendered programmatically from a graph description, and its edited counterpart is rendered independently from the corresponding edited graph through the same code path. The intended difference between the two inputs is therefore exactly one graph edit rather than a manual pixel-level modification.

Candidate-edge matching is verified on the rendered stimuli before activation extraction. The two candidate edges must agree in source icon, target icon, predicate, drawn length, orientation, and target out-degree. Items violating any construction constraint are rejected rather than retained and corrected statistically afterward.

These checks are identification constraints rather than aesthetic filters: their purpose is to prevent the nominal graph-role label from being predictable from the rendered candidate edge or its immediate neighborhood. Concretely, appearance and entity identity are matched, with both candidates using identical endpoint types, labels, and rendering parameters; local geometry is matched in orientation, length, endpoint configuration, and degree; relation type is held fixed within each comparison; global chain structure is crossed with role, so that total lane length cannot determine the label; and position is entered into the statistical baseline rather than assumed away by randomization.

Distance-controlled stimuli. In the distance-controlled stimulus set used in Section 4.3, the candidate edge occupies the same rendered box,

$$(157.9, 140.3, 333.6, 228.4)$$

pixels in every condition. The candidate-to-role-arrow gaps agree to within 0.05 pixels across all 112 corresponding items within each intended distance condition.

One mirror arm of the distance intervention was constructed and rendered but excluded before activation extraction. Compressing the naturally longer $n = 5$ layout to approximately 263 pixels produces node-label collisions. That manipulation would therefore confound physical distance with a change in rendered visual content, so it is not used in the reported analysis.

D REMOTE-EVIDENCE IDENTIFICATION EXPERIMENT

Ten arms, each holding the graph, the hop count, the entity identities, the edge directions and the rendered candidate edge fixed and changing only the remote role-defining arrow. They were built and pre-registered before extraction (Amendment 15), across three extractions of 56 base items $\times$ 6 layout permutations each, at $s = 0.2$ under arrow reversal on Qwen-Image.

Where the arrow is drawn. The gap between the candidate edge and the role-defining arrow is set by stretching the lane, at 262.9, 314.1, 365.3 and 416.5 pixels with the node span fixed at 0.20, and at 238.0 and 442.8 pixels with the span at 0.15. The far/wide arm sits at 416.5 rather than 442.8 pixels because matching that distance at a 0.20 span requires a node centre at 0.97, whose icon box leaves the 1024-pixel canvas. `_trim`'s minimum-length floor clamps the drawn arrow to 89.6–91.0 pixels in every arm, so arrow length is matched except where it is the manipulation.

How the arrow is drawn. Three families. A plate-grey patch (`meta["occluders"]`, drawn last) covers a solved fraction of the arrowhead, with exposure solved by bisection on the rendered image so each arm hits its target to ± 0.02 in every template; `disp33` places the identical patch just past the arrow tip as the control for the patch itself. `long` stretches the role arrow's shaft from 91 to 221 pixels at fixed endpoints. Tightening the node span to 0.15 crowds the target icon against the arrowhead, and `occ15top` redraws the role edge above the icons at that span as the control for the crowding.

Head exposure, measured rather than inferred. `rolegraphs.head_exposure` masks the arrowhead triangle against the finished image and returns the inked fraction, which is what the model sees; this replaces a geometric calculation that treated an icon as a solid box.

Table 1: Measured arrowhead exposure per arm, 56 items.

arm	node span	mean	sd	range
263 px	0.20	0.917	0.023	0.81–0.93
416 px	0.20	0.921	0.029	0.78–0.94
238 px	0.15	0.792	0.179	0.31–0.95
443 px	0.15	0.786	0.183	0.29–0.95

The tight-span arms are not uniformly occluded: several templates draw sparse glyphs that hide almost nothing. Splitting the 443-pixel extraction by template, post-hoc, at block 38:

Reproductions inside the design. Three arms are positive controls that reproduce the frozen $n = 4$ cell rather than testing anything: the uncovered patch arm reads +0.9298, +0.9349, +0.4980, +0.4041, +0.7892 across the band against the frozen +0.9296, +0.9352, +0.4982, +0.4040, +0.7891; the wide-span crowding arm reproduces the same five values; and the 263-pixel arm is bit-identical across the three extractions that contain it. The tight-span arm and its re-extraction after a layout-dispatch fix agree to three decimals (+0.0157 against +0.0158 at block 38).

Table 2: Exposure and role ΔS by template at block 38. The three templates whose arrowhead is essentially unoccluded null as hard as the rest, which is why the exposure account is withdrawn. Post-hoc split of an existing extraction, not a pre-registered comparison.

template	exposure at 443 px	ΔS at 443 px	ΔS at 263 px
growth_chain	0.926	+0.0176	+0.9892
light_response	0.926	+0.0146	+0.9561
feedback	0.918	+0.0104	+0.9684
circuit	0.871	+0.0132	+0.9760
food_web	0.735	+0.0083	+0.9388
heating	0.576	−0.0095	+0.8677
matter_cycle	0.550	−0.0015	+0.8496

Table 3: **Ten arms across the Qwen-Image late-middle band**, role ΔS / accuracy, six layout permutations. Rows above the rule change only where the arrow is drawn; rows below change how it is drawn. Every gate passes. * marks a cell whose label-shuffle interval excludes zero; the flags cluster in the patched and crowded arms, with offsets of −0.003 to −0.009 bits against effects of +0.07 to +0.09, and one falls on the unmodified baseline at block 41, where the offset is −0.008 against an estimate of +0.499. Every offset is negative, so a true zero would read low rather than high here, and the causal claim rests on Table 11 rather than on any of these cells.

arm	gap / exposure	35	38	41	44	47
none	263 / 0.92	+0.930 / .987	+0.935 / .993	+0.499* / .831	+0.404 / .813	+0.789 / .949
moved farther	416 / 0.92	+0.967 / 1.00	+0.969 / 1.00	+0.954 / .998	+0.984 / 1.00	+0.980 / 1.00
patch past tip	263 / 0.92	+0.696 / .915	+0.934 / .995	+0.146* / .691	+0.173* / .693	+0.458* / .827
redrawn on top	263 / 0.95	+0.930 / .985	+0.950 / 1.00	+0.723 / .927	+0.489 / .846	+0.829 / .961
head 35% covered	263 / 0.65	+0.095* / .644	+0.094* / .590	+0.012* / .544	+0.017* / .582	+0.043* / .611
head 68% covered	263 / 0.32	+0.061* / .611	+0.081* / .608	+0.007* / .555	+0.019 / .594	+0.020* / .564
head fully covered	263 / 0.00	+0.045* / .586	+0.071* / .612	+0.010* / .568	+0.016* / .580	+0.013* / .539
shaft 91 → 221 px	264 / 0.93	+0.075* / .605	+0.019* / .547	+0.005* / .537	+0.003 / .528	+0.011* / .548
span tightened	238 / 0.79	+0.008 / .545	+0.016 / .559	+0.019* / .590	+0.004 / .520	+0.022 / .583
span tightened, far	443 / 0.79	+0.006 / .531	+0.008 / .548	+0.017* / .557	+0.007* / .537	+0.024 / .586
span tightened, top	238 / 0.95	+0.024* / .563	+0.021* / .568	+0.017* / .571	+0.006* / .528	+0.028* / .607

Token composition of the arrowhead region. A token is a 16-pixel cell. Averaged over the tokens the arrowhead falls into, the share that is arrow ink is 0.375–0.398 in the positive arms, 0.443 in the shaft-stretched arm and 0.025 in the fully covered arm, so it does not order the arms (Spearman +0.175 against ΔS). Icon ink does separate the tight-span arms (0.194 and 0.207) from every positive arm (0.058–0.075), with the redrawn-on-top arm intermediate at 0.135, and says nothing about the shaft-stretched arm, whose head tokens are the cleanest in the set at 0.038. We report it as a candidate sub-mechanism covering two of the three failure modes, not as a mechanism.

Topology-summary diagnostic. Conditioning on post-edit topology summaries does not reduce the frozen effect, +0.380 bits against +0.382 without them. But graph role is not decodable from those summaries either, accuracy exactly 0.500, because reversing the first edge of either congruent lane leaves the relevant global counts unchanged. It is a valid negative result for those summaries and not the mediation test it was intended to be.

E A SECOND STRUCTURAL FAMILY

Every positive in this study comes from one topological role, *extended vs terminal*. This family, pre-registered as Amendment 19 before any extraction, asks whether the apparent signal appears for a different role built by the same means.

Construction. Seven templates $\times$ eight variants = 56 items, matched in count to the 56 $n = 4$ items every arm of Appendix D was built from. Each candidate lane is a five-node row: spine $L_0 \rightarrow L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_4$ with the candidate edge $L_0 \rightarrow L_1$ at the same canvas positions as the primary family, plus one bowed remote arrow between L_2 and L_4 whose direction is the entire

manipulation: $L_4 \rightarrow L_2$ closes the cycle $L_2 \rightarrow L_3 \rightarrow L_4 \rightarrow L_2$, while $L_2 \rightarrow L_4$ bypasses L_3 and leaves a directed acyclic graph. The arrow is bowed because a branch point requires a node with two successors, which a single row cannot draw without collisions, and it is attached at L_2 and L_4 so that *both* arrowheads stay remote; attaching it at L_1 would place a head inside the readout mask and make the role locally visible, which is the confound the design exists to avoid. The control lane uses the same reserved entities as everywhere else.

What is matched, verified on all 56 rendered items before extraction. Nodes and edges per lane, five and five; the candidate edge’s position; the reachable set downstream of each candidate’s target, three nodes; the longest simple path from it, three; and drawn ink per lane, whose mismatch across the 56 items has a maximum of 0.91% and a mean of 0.35%. The primary family’s role label is *extended* for both lanes, so neither it nor chain length can separate the pair. One quantity is not matched and is reported rather than hidden: a seen-guarded longest-walk statistic differs by one step between the lanes, because the walk traverses the closing edge before it stops. Reachability and path length are equal, so this is a property of that statistic and not a structural asymmetry.

Table 4: The second family at the five band blocks, Qwen-Image reversal at $s = 0.2$, six layout permutations, 2016 rows. The gate passes at every block, so every cell is readable, and the role estimates lie inside $\pm\delta$ by two orders of magnitude. * marks a cell whose label-shuffle interval excludes zero; those offsets exceed the estimates and point negative.

block	validity ΔS	val. acc / majority	role ΔS	CI ₉₀	role acc
35	+0.5166	0.836 / 0.667	+0.0007*	[+0.0002, +0.0011]	0.511
38	+0.5546	0.876 / 0.667	+0.0020*	[+0.0001, +0.0041]	0.519
41	+0.4621	0.815 / 0.667	+0.0009*	[−0.0007, +0.0027]	0.513
44	+0.4306	0.818 / 0.667	−0.0011	[−0.0020, −0.0002]	0.496
47	+0.4458	0.816 / 0.667	−0.0008	[−0.0017, +0.0001]	0.514

Reading, and what it does not settle. The validity contrast is weaker than the primary family’s, +0.431 to +0.555 bits at accuracies of 0.815 to 0.876, against roughly +0.9 bits at accuracy 0.95 or better there, so this family is the less sensitive instrument and the null is a null under it. The shuffle flags at blocks 35, 38 and 41 carry offsets of −0.0023 to −0.0064 bits, larger in magnitude than the estimates and negative in sign, so a true zero would read slightly positive here and the displacement plausibly *is* the estimate; the two shuffle-clean blocks read −0.001. Nothing in this measurement identifies why the role is not decodable: the bow, the two arrowhead positions, and the absence of any signal are not separated, and the pre-registered rule for a negligible outcome ran no further intervention.

F FULL PER-CELL RESULTS

model	edit	s	depths	validity	role ΔS	status
Qwen-Image	deletion	0.2	8	pass, +0.915	$+3 \times 10^{-5}$	negligible
Qwen-Image	deletion	0.4	4	pass	-4×10^{-4}	negligible
SD3.5	deletion	0.4	2 of 4	pass, +0.908	$+3 \times 10^{-6}$	negligible
SD3.5	deletion	0.2	2 of 4	pass	-4×10^{-4}	negligible
SD3.5	reversal	0.2	8	pass	$\leq +0.019$	negligible
Qwen-Image	reversal	0.2	8	pass	$\leq +0.024$	negligible

Table 5: **Final controlled results.** Role ΔS is interpreted against the pre-specified $\delta = 0.10$ -bit equivalence threshold and reported only where the construction-validity gate passes. The two early SD3.5 deletion cells that fail the gate at accuracies 0.667 and 0.633 are treated as unreadable rather than as evidence for a null. The Qwen-Image reversal row reports the distance-controlled condition of Section 4.3. Full per-cell estimates, intervals, and flags are given in the paragraphs that follow.

Cells are labeled CLEAN, SHUFFLE-BIASED when the label-shuffle interval excludes zero, and GATE-FAILED when construction-validity accuracy does not exceed the majority-class rate. A GATE-FAILED cell is treated as unreadable rather than as evidence for a negligible effect. A SHUFFLE-BIASED cell may be readable but is not used as clean confirmatory evidence.

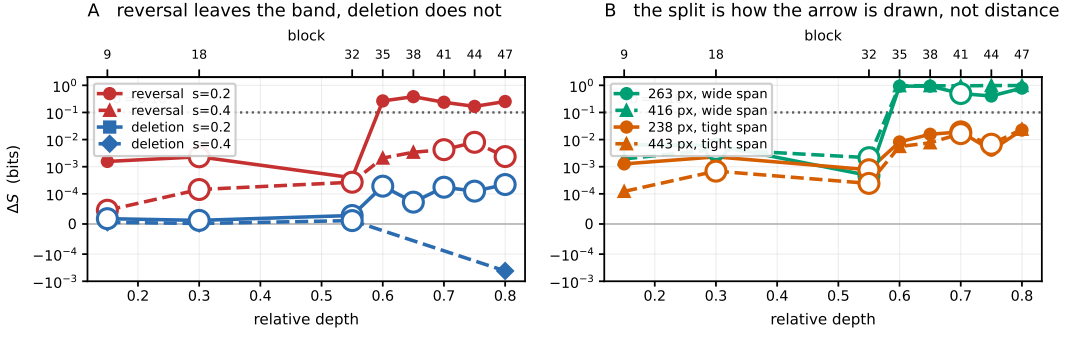


Figure 5: **A depth-localized band, and what separates the cells inside it.** Role ΔS across transformer blocks against the equivalence region $\pm\delta$; open markers are cells whose label-shuffle interval excludes zero and which are therefore not interpreted. (A) Qwen-Image reversal at $s = 0.2$ leaves the region across late-middle blocks, while deletion and the higher-strength condition stay negligible wherever readable. (B) The same cells split by how the role-defining arrow is drawn, at the wide and the tight node span, crossed with its drawn distance: colour is the node span and marker is distance. The two wide-span arms sit together above the bound at every band block whether the arrow is 263 or 416 pixels away, while the two tight-span arms sit together inside it, at 238 and at 443 pixels. Stable Diffusion 3.5 Large, whose profile has the same shape an order of magnitude smaller and inside the bound everywhere, is Figure 6 in Appendix F.

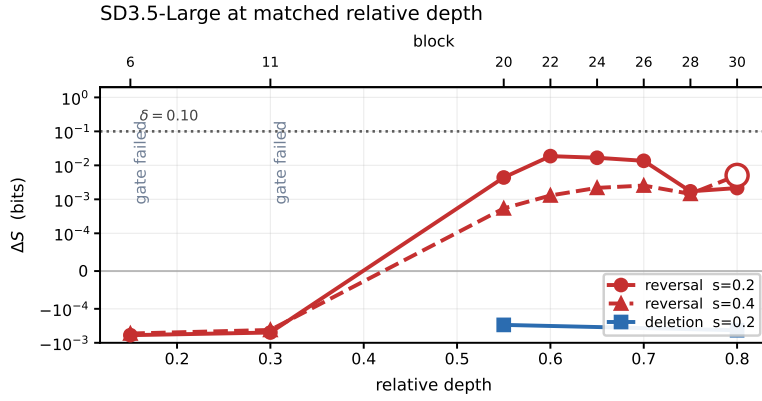


Figure 6: **Stable Diffusion 3.5 Large at matched relative depth.** The reversal profile has the same shape as the Qwen-Image band of Figure 5 but is roughly an order of magnitude smaller and stays inside the equivalence bound at every readable depth, with a maximum of $+0.019$ bits. Two early deletion cells fail the construction-validity gate and are marked rather than plotted. We report this as found and do not characterize it further.

Of the sixteen signed cells in the confirmatory grid, eight are CLEAN. The remaining cells are retained in the complete result record but are not interpreted as clean confirmatory evidence when their validity or shuffle criteria fail.

Qwen-Image, deletion. Across eight sampled blocks at $s = 0.2$, role estimates lie between 1.2×10^{-5} and 2.2×10^{-4} bits, with role accuracy 0.499–0.511 and validity accuracy 1.000.

At $s = 0.4$, the four confirmatory blocks yield $+5 \times 10^{-6}$, $+2 \times 10^{-6}$, $+1.1 \times 10^{-5}$, and -4.2×10^{-4} bits, with role accuracy 0.498–0.503. The block-32 cell is SHUFFLE-BIASED.

Qwen-Image, reversal. At $s = 0.2$, the confirmatory grid contains the late-depth apparent positive characterized in Section 4.3. Dense sampling shows that this effect extends across a contiguous late-middle band rather than arising from a single block.

The distance-controlled intervention reported in Appendix D reduces the corresponding band-block estimates to

$$+0.006, +0.007, +0.017, +0.007, +0.024$$

bits, all inside the pre-specified equivalence region. At $s = 0.4$, the corresponding readable cells remain within the equivalence region.

Stable Diffusion 3.5 Large, deletion. At $s = 0.2$, blocks 6 and 11 fail their construction-validity gate at accuracies 0.667 and 0.633, respectively. Blocks 20 and 30 yield -2.97×10^{-4} and -4.27×10^{-4} bits.

The same early-depth validity failures are also present at $s = 0.4$. These cells are reported as unreadable rather than as evidence for a null.

Stable Diffusion 3.5 Large, reversal. At $s = 0.2$, the eight sampled blocks yield

$$\begin{aligned} &-5.96 \times 10^{-4}, -4.99 \times 10^{-4}, +0.0044, +0.0187, \\ &+0.0167, +0.0136, +0.0017, +0.0021 \end{aligned}$$

bits. All lie within δ , with clean label-shuffle controls and validity accuracy between 0.817 and 0.888.

At $s = 0.4$, the same region remains flat. By the pre-specified criterion, the Qwen-Image-like elevated band is therefore absent in this model.

A qualitatively similar rise appears near the relative depths corresponding to the Qwen-Image band, but at approximately one fourteenth of its magnitude. This pattern was not pre-registered, no outcome interpretation was specified for it in advance, and we report it only as an uncharacterized observation.

G PROBE AND CONTROL DIAGNOSTICS

The analyses in this section evaluate diagnostics that were introduced to interpret or challenge apparent probe signals. They are reported separately from the controlled confirmatory result because their purpose is to establish whether the diagnostic itself supports the inference it was intended to make.

G.1 SHORTCUT REDESIGNS AND ESTIMATOR DIAGNOSTICS

This subsection gives the full version of the cases summarized in Figure 2 and Section 4.2, together with the two criterion failures stated more briefly there.

Entity identity. Before endpoint identity was controlled by construction, a probe separated the two candidate roles at $\Delta S = +0.539$ bits, $\text{CI}_{90} [+0.229, +0.837]$. Restricting the analysis to the one stimulus family whose candidate edges use identical endpoint icons and labels reduces the estimate to -0.001 bits, $\text{CI}_{90} [-0.303, +0.302]$. The interval is wide on only 32 items, so this subset does not establish a negligible effect; it does, however, exclude an effect of the magnitude originally observed. The initial signal was therefore explained by the identities of the objects incident to the edited edge. Notably, the construction-validity gate and the label-shuffle control both pass in the confounded design. Neither can identify this shortcut, because both are compatible with a readout that predicts the nominal role label from endpoint identity. Successful controls do not substitute for stimulus design that breaks specific nuisance correlations.

Global chain length, amplified by decoder capacity. After the identity shortcut was removed, a nonlinear probe produced an even larger apparent role signal. The MLP recovered $\Delta S = +0.556$ bits, $\text{CI}_{90} [+0.530, +0.582]$, while a linear probe on the same rows recovered only $+0.006$ bits, which initially suggested that graph role might be encoded nonlinearly. Crossing graph role with total lane length by construction removes the effect: the same MLP, at the same block and timestep, falls to $\Delta S = +0.00002$ bits. The candidate arrows remain locally matched; the exploited variable is

variable	within del.	within rev.	rev.→del.	del.→rev.	trained on both
graph role	0.499	0.724	0.527	0.502	0.503 / 0.560
validity contrast	1.000	0.836	0.546	0.667	1.000 / 0.821
edit position	0.905	1.000	0.667	0.667	0.871 / 0.913

Table 6: **Failure to transfer across edit type is not diagnostic of absence.** Probe accuracy at the deep Qwen-Image condition. The validity contrast and edit position serve as positive controls: both are present and decodable within edit type, yet cross-condition transfer falls to 0.667 or below. Joint training recovers them, indicating condition-specific encoding directions rather than absence of information.

instead global chain length, which is nevertheless recoverable at the local edit site. Greater decoder capacity therefore increased predictive performance by making a nuisance variable accessible, rather than by providing stronger evidence for the intended representation.

Estimator bias. An early confirmatory analysis returned $\Delta S = -0.186$ bits, $CI_{90} [-0.204, -0.168]$, on a cell whose validity gate passed decisively. The estimate was precise and excluded zero, but the corresponding label-shuffle distribution was centered at approximately -0.099 bits, so the estimator’s no-information point was displaced from zero and the equivalence test was uninterpretable. This failure reflects bias rather than insufficient sample size: more data would narrow the confidence interval around the displaced reference point rather than correct it. Fitting a separate probe within each item produced this behavior, so all reported results use the pooled estimator.

A validity score that certifies too little. The construction-validity contrast is a between-lane discrimination, whereas the target role contrast is within-item. A readout built directly on the edited image separates candidate from control at 0.833 accuracy, above the corresponding 0.667 majority rate, while predicting role at exactly 0.500. More importantly, on the reversal stimuli an image readout obtains $+0.128$ bits on the validity contrast while its accuracy remains exactly at the majority rate. An information score can therefore appear favorable even when the readout does not discriminate beyond the relevant baseline. We consequently require the validity gate to satisfy both the information criterion and accuracy above the majority-class rate.

G.2 SHUFFLE FLAGS IN THE PATCHED ARMS

Most cells in the arms with a patch over the arrowhead have label-shuffle intervals excluding zero, so by our own rule they are not clean confirmatory evidence, and Table 3 marks each one. The offsets are -0.003 to -0.009 bits against effects of $+0.07$ to $+0.09$: too small to manufacture the collapse, and pointing the wrong way. The causal claim rests on the masking conditions of Section 4.4 instead, where every cell is shuffle-clean and the manipulation is on the model rather than on the image.

G.3 CROSS-EDIT TRANSFER

We initially considered cross-edit invariance as a potential test of graph-role representation. Because graph role is unchanged when the candidate edge is deleted rather than reversed, a representation encoding role might be expected to support transfer across edit type.

The graph-role probe transfers from reversal to deletion at only 0.527 accuracy compared with 0.724 within reversal. Considered alone, this result would appear to argue against a shared graph-role representation.

The positive controls show why that interpretation is not justified. The validity contrast falls from 1.000 within deletion to 0.546 across edit types, while edit position falls from 0.905 to 0.667. Both quantities are known to be present under both interventions.

Training a single probe jointly on both edit types recovers these variables, indicating that their information is available under both conditions but represented in condition-specific directions.

Cross-edit failure is therefore not diagnostic of absence in this setting. Before a transfer or invariance criterion can be used to reject a target representation, the same readout should first demonstrate successful transport of variables known to survive the transformation.

G.4 THE LAYOUT-COUNT AUDIT

The masking conditions are probed on three of the six layout permutations, so that three extractions fit the compute available. Their unmasked control reads +0.7301 at block 38 where the frozen cell reads +0.9351, a discrepancy that has to be explained before the masking table can be printed at all, because the obvious reading is that the wrapper perturbs attention even with no keys removed.

It does not. The estimator conditions on position and holds layouts out as folds, so three permutations leave a different position marginal: majority-class accuracy rises from 0.667 to 0.778 and validity ΔS falls at every block, which leaves less headroom above the baseline and a smaller ΔS . The shift is not uniform, and that is the tell: blocks 44 and 47 go up, whereas a corrupted forward pass attenuates everywhere. Re-probing the frozen extraction on exactly those three permutations, from features already on disk and with no GPU, gives +0.6145, +0.7299, +0.1744, +0.4853 and +0.9431 at blocks 35, 38, 41, 44 and 47 against the masking run’s empty-mask column of +0.6143, +0.7301, +0.1743, +0.4850 and +0.9432: a maximum discrepancy of 0.0004 bits, with validity agreeing to four decimals (+0.5998 both, accuracy 0.970 both, majority 0.778 both).

Table 7: **The two estimators side by side**, Qwen-Image reversal at $s = 0.2$, the 263-pixel arm, 56 base items. The left half is the frozen six-layout extraction every number outside the masking experiments is quoted from; the right half is the same features re-probed on perm0, perm2 and perm4, which is the estimator the masking tables use. Role ΔS falls at blocks 35, 38 and 41 and rises at 44 and 47, which a corrupted forward pass could not produce; validity ΔS falls everywhere because the higher majority leaves less headroom, while validity accuracy rises for the same reason. This table is the conversion between the two, and the rest of the paper points here rather than restating it.

block	six layouts, majority 0.667			three layouts, majority 0.778		
	role ΔS / acc	val ΔS	val acc	role ΔS / acc	val ΔS	val acc
35	+0.9298 / .987	+0.717	.943	+0.6145 / .905	+0.571	.959
38	+0.9351 / .993	+0.784	.969	+0.7299 / .938	+0.600	.970
41	+0.4985 / .831	+0.651	.915	+0.1744 / .710	+0.562	.948
44	+0.4038 / .813	+0.617	.894	+0.4853 / .854	+0.497	.933
47	+0.7891 / .949	+0.620	.896	+0.9431 / .999	+0.466	.919

Three things follow. The empty mask is a genuine pass-through. The matched baseline for the masking table is the three-layout re-probe rather than the frozen six-layout number, and the paper quotes it that way. And this is a fourth reproduction of the 263-pixel cell, the only one crossing code paths, comparing an extraction with the masking wrapper installed against one without it. One thing the reduced set does worse: its label-shuffle control flags far more often outside the band and in null arms. At blocks 9, 18 and 32, sixteen of the eighteen Qwen-Image masking cells flag; inside the band the 263-pixel panel is clean throughout, while four of the fifteen shaft-stretched cells flag with offsets of -0.008 to -0.013 against effects of ± 0.005 . Every one of those offsets is negative. Six permutations remain the estimator everywhere else in the paper, and the reduced set appears only where conditions are compared with each other. Two cells fitted on different numbers of layouts are different measurements, and we label which is which wherever both appear.

G.5 ADDITIONAL PROBE VARIANTS

The MLP replication reuses the same data partitions, activation features, and control logic as the linear analysis. Its regularization is calibrated using the label-shuffle control rather than validation accuracy, because the default setting drives the probe toward a constant predictor; the reported cells use the calibrated penalty $\alpha = 10$.

That calibration is also what rules out the constant predictor as an explanation of the null. At $\alpha = 10$, on the same rows and folds, the decoder classifies the validity contrast at accuracy 1.000 against a

readout	validity	role ΔS	interpretation
linear, local	+0.915/ + 0.908	$+3 \times 10^{-5} / + 3 \times 10^{-6}$	primary
MLP, local	+0.907/ + 0.898	$-1 \times 10^{-5} / + 2 \times 10^{-5}$	clean shuffle
pairwise endpoints	weak	+0.0046	3 of 6 fail interpretation criteria
whole sequence	pass (Qwen)	+0.205 (frozen band cell)	gate fails for SD3.5
AI2D-RST, 349 diagrams	—	−0.030	corroborating

Table 8: **Additional probe variants.** Paired values are Qwen-Image / Stable Diffusion 3.5 Large where applicable. A small role estimate is interpreted only when the corresponding validity and label-shuffle criteria are well behaved.

0.667 majority rate, with $\Delta S = +0.907$ for Qwen-Image and +0.898 for Stable Diffusion 3.5 Large, while role accuracy is exactly 0.500 with ΔS of -9.5×10^{-6} and $+1.7 \times 10^{-5}$. A probe that had collapsed to predicting one class would fail the validity contrast as well; this one does not. Despite its greater capacity, the MLP therefore preserves the controlled pattern: validity high, graph-role information negligible.

The pairwise endpoint readout pools source-node and target-node tokens separately and supplies both vectors, their difference, and their elementwise product. Although the resulting role estimate is small, three of its six cells have label-shuffle intervals excluding zero and its positive-control behavior is weak. We therefore treat this analysis as inconclusive rather than confirmatory.

A whole-sequence readout is usable for Qwen-Image but fails its construction-validity gate for Stable Diffusion 3.5 Large. Accordingly, we do not use the whole-sequence analysis to support an architecture-independent claim.

Text-conditioning diagnostic. A separate analysis asks whether the text-conditioning representation distinguishes a reversed relation from a relation-preserving reordering. Under mean-pooled signed features, the preserved control separates more strongly than the reversal in both models, while the direction of the effect changes under sequence-level features.

We therefore treat this analysis as unsettled and do not use it to support the central representational claim.

H ATTENTION MEASUREMENT AND THE KEY-MASKING INTERVENTION

Capture. For the image tokens the readout pools over, we record the attention mass their queries place on the role-defining arrowheads, those arrows’ shafts, the candidate region, the rest of the image and the text stream. The model’s own attention function is called for the output; the reported weights are recomputed for those query rows alone, so the forward pass is numerically unchanged and the full token-by-token matrix is never formed. Qwen-Image concatenates the joint sequence as [text, image] and Stable Diffusion 3.5 Large as [image, text]. Values are enrichment over a uniform reference, mass divided by the region’s share of tokens; the arrowhead region is 13–20 of 4096 tokens, so the ratio is noisy. Predictions for every arm were registered before the capture (Amendment 16).

Scaling the capture $21\times$, to $56 \text{ items} \times 3 \text{ layouts}$ on the four arms that carry the distance contrast, keeps the split at four of the five band blocks and reverses it at blocks 18 and 32. It is weakest at block 38, 1.77 ± 0.24 for the positive arms against 1.61 ± 0.32 for the null ones, the reference block for the rendering comparisons, where ΔS differs by two orders of magnitude.

The shaft-stretched arm at the same scale, unregistered. The four arms above were chosen before the capture ran. The shaft-stretched arm was not among them, which left the most striking cell of Table 9 resting on eight items. We re-ran it on the identical protocol, the same 56 items and the same perm0, perm2 and perm4 layouts, 168 passes. This analysis was run after the eight-item numbers were read and no outcome was committed for it in advance; it is stage 13 of Table 12 and we treat it as exploratory. Enrichment is attention mass on the arrowhead region divided by that region’s share of the 4096 image tokens, and the region spans 15 to 20 tokens across these arms, so the denominator differs between rows and the ratio stays noisy.

Table 9: **Head-region attention enrichment**, 8 base items $\times$ 1 layout per arm, so that all ten arms are measured on one protocol. Positive arms separate from null arms only from block 35, the depth at which the band begins. Two null arms carry enrichment above the tight-span nulls while reading null, and on this protocol the shaft-stretched arm exceeds every positive arm at block 35. That last reading does not survive the scale-up in Table 10 and is withdrawn there.

arm	role ΔS	blk 18	blk 32	blk 35	blk 38	band mean
263 px	+0.9351	1.31	1.98	2.43	1.93	2.28
416 px	+0.9687	1.06	1.59	2.15	1.68	2.25
patch past tip	+0.9342	1.39	2.02	2.29	1.92	2.18
shaft 91 $\rightarrow$ 221 px	+0.0192	1.13	1.85	2.50	2.06	1.83
span tightened, top	+0.0207	1.33	1.61	1.43	1.62	1.54
span tightened	+0.0157	1.28	1.77	1.33	1.36	1.28
span tightened, far	+0.0076	1.15	1.66	1.36	1.45	1.38
head 35% covered	+0.0943	1.47	2.26	1.67	1.24	1.35
head 68% covered	+0.0815	1.10	1.78	1.11	0.70	0.77
head fully covered	+0.0711	1.01	1.22	0.88	0.54	0.56

Table 10: **Head-region enrichment at 56 items $\times$ 3 layouts**, mean $\pm$ standard deviation across the 168 passes of each arm. The first four rows are the registered distance-contrast arms; the last is the unregistered scale-up of the shaft-stretched arm. At blocks 35 and 38 the shaft-stretched arm has enrichment comparable to the positive arms despite much lower decodability. At block 38 its mean enrichment exceeds the 416-pixel arm’s, whereas at blocks 41 and 47 the ordering is the expected one. This exploratory comparison describes the ordering of arm means, not a test of individual tokens’ importance.

arm	role ΔS	blk 35	blk 38	blk 41	blk 47
263 px	+0.935	2.45 $\pm$ 0.36	1.94 $\pm$ 0.23	2.56 $\pm$ 0.28	3.29 $\pm$ 0.35
416 px	+0.969	2.17 $\pm$ 0.34	1.63 $\pm$ 0.25	2.81 $\pm$ 0.47	3.53 $\pm$ 0.42
span tightened	+0.016	1.53 $\pm$ 0.38	1.60 $\pm$ 0.33	1.77 $\pm$ 0.43	1.92 $\pm$ 0.53
span tightened, far	+0.008	1.62 $\pm$ 0.40	1.64 $\pm$ 0.31	2.20 $\pm$ 0.64	2.34 $\pm$ 0.72
shaft 91 $\rightarrow$ 221 px	+0.019	2.15 $\pm$ 0.22	1.88 $\pm$ 0.16	1.49 $\pm$ 0.15	1.55 $\pm$ 0.22

The eight-item reading of 2.50 at block 35 does not replicate: the same cell reads 2.15 ± 0.22 here, against 2.17 ± 0.34 for the 416-pixel arm, so the claim that this arm carries the strongest routing we measure is withdrawn. What the scale-up does establish is firmer than what it removes. At block 38 the arm reads 1.88 ± 0.16 on $+0.019$ bits against 1.63 ± 0.25 on $+0.969$ bits, an inversion rather than an absence of ordering, and the three arms within 0.011 bits of one another span 1.60 to 1.88. Attention on the candidate region is 0.3140 here against 0.3136–0.3174 across the four registered arms, so this is not a global shift in how much those query rows attend to anything. Capturing the two candidate-edited images as well, so the measure matches what the probe reads, does not order the arms either: the difference in head enrichment between the two edits is 0.022 and 0.029 in the positive arms against 0.020 and 0.008 in the null ones. Attention on the candidate region itself agrees across arms to two decimals in both models, 0.3136–0.3174 at block 38 in Qwen-Image, so none of this is a global shift in how much those tokens attend to anything.

Stable Diffusion 3.5 Large. The same structure appears and is stronger. At matched relative depth 0.65, which is block 24 of its 38 and block 38 of Qwen-Image’s 60, SD3.5 reaches enrichment 4.12 on the wide-span stimuli where Qwen-Image reaches 1.93, and 2.03 on the tight-span stimuli. Its peak role estimate is $+0.021$ bits against Qwen-Image’s $+0.943$ in the matched masking extraction. It therefore routes $2.1\times$ harder onto the region for $45\times$ less decodable information, through a path the next paragraph shows is causally necessary in both.

Intervention. Keys covering the role-defining arrowheads are removed from the candidate-edge queries’ softmax at every block and only those output rows are replaced. The control removes a comparable number of keys covering the control lane’s arrowhead (≈ 12 against ≈ 15 tokens). All

three conditions run through the identical code path; with no target set the wrapper returns the model’s own attention output with no recompute and no splice.

Table 11: **Key masking, all conditions and blocks**, role ΔS / accuracy, three layout permutations in every panel, so the position baseline is a 0.778 majority rather than 0.667 and only the contrast between rows is read. Qwen-Image and the shaft-stretched arm are at $s = 0.2$ under reversal; SD3.5 is at $s = 0.4$. * marks a cell whose label-shuffle interval excludes zero. Every printed cell of the 263-pixel Qwen-Image panel and every printed SD3.5 cell is clean. Four of the fifteen printed shaft-stretched cells are flagged: the empty mask at block 47, the head cut at blocks 35 and 47, and the control cut at block 35. Their offsets run -0.008 to -0.013 bits and are all negative, so each displaces an estimate downward rather than upward.

	keys removed	35	38	41	44	47
Qwen, 263 px	none	+0.6143 / .909	+0.7301 / .938	+0.1743 / .711	+0.4850 / .856	+0.9432 / .999
	arrowheads	+0.0034 / .540	+0.0046 / .525	+0.0031 / .527	-0.0017 / .503	-0.0048 / .476
	control lane	+0.6206 / .874	+0.6357 / .866	+0.1458 / .680	+0.4363 / .820	+0.9459 / 1.00
Qwen, shaft	none	+0.0460 / .631	+0.0046 / .516	+0.0008 / .501	-0.0039 / .499	+0.0056* / .551
	arrowheads	-0.0021* / .510	-0.0035 / .499	-0.0059 / .484	-0.0057 / .476	-0.0001* / .518
	control lane	+0.0990* / .626	+0.0080 / .542	+0.0003 / .510	-0.0029 / .519	+0.0057 / .539
		22	24	26	28	30
SD3.5, 263 px	none	+0.0012 / .516	+0.0092 / .552	+0.0136 / .558	+0.0110 / .564	+0.0212 / .571
	arrowheads	-0.0056 / .494	-0.0011 / .509	-0.0004 / .488	-0.0016 / .491	-0.0005 / .501
	control lane	+0.0017 / .536	+0.0131 / .577	+0.0144 / .567	+0.0131 / .564	+0.0168 / .552

Validity is undisturbed throughout: accuracy 0.951–0.980 across the three Qwen-Image conditions at block 38, 0.880–0.934 across the shaft-stretched conditions, and 0.821–0.978 across the SD3.5 conditions, all against a 0.778 majority, with SD3.5’s validity slightly higher under the head cut at block 26 than without it.

Two observations on the shaft-stretched arm were not among the pre-registered outcomes and are reported as unregistered. The small elevation that arm carries at block 35, +0.0460 [+0.0294, +0.0632] and inside the equivalence bound, is removed to chance by the head cut, -0.0021 [-0.0087, +0.0043], with non-overlapping intervals, so the null is weaker at that block than a single peak-block quote would suggest. And the control cut *raises* it to +0.0990 [+0.0832, +0.1147], landing on the equivalence bound. The pre-registration named the control cut *collapsing* as the outcome that would invalidate the intervention; an increase was not among the branches. Softmax competition is the natural account, since masking keys renormalizes the remaining mass toward the role arrowheads, and it is untested. It strengthens rather than weakens the specificity claim: a key-count account predicts both cuts degrade the estimate, and here two comparable cuts move it in opposite directions in the same arm, at the same block, through the same code path. A 0.099-bit effect at one block of eight, in an arm null everywhere else, is a lead worth one experiment rather than a result.

The comparison of masking conditions against the six-layout estimates elsewhere in the paper is audited in Appendix G.4.

I CORROBORATION ON NATURALLY OCCURRING DIAGRAMS

Candidate arrow pairs in AI2D-RST are selected from naturally occurring textbook diagrams and matched on arrow area and orientation, which is the closest matching permitted by the corpus.

Unlike in the controlled synthetic stimuli, endpoint identity, label content, local geometry, and surrounding graph context cannot be matched exactly. Accordingly, the real-diagram experiment does not reproduce the full identification guarantees of the synthetic construction.

The resulting analysis yields

$$\Delta S = -0.030, \quad \text{CI}_{90} = [-0.040, -0.019],$$

with a label-shuffle estimate of approximately -0.002 . Role accuracy is 0.492, compared with 0.585 for the position-only baseline.

We therefore interpret this experiment as corroborating evidence that the near-zero controlled result is not unique to programmatically rendered stimuli, rather than as an independent confirmation of the synthetic identification result.

J ANALYSIS PROVENANCE AND PRE-REGISTRATIONS

A construction bug, and what it did not touch. Two arms of the occlusion sweep were built at one node spacing and extracted at another: the extractor renders from the layout-variant function rather than from an item’s own base layout, and that function had no branch for the new arms, so both reverted to the wide spacing. The pair was re-extracted after the fix, which now asserts that the variants agree with the item’s base layout. The error is visible in the record rather than hidden by it: the reverted arms reproduced the wide-spacing cell to four decimals — $+0.9296/ +0.9351/ +0.4984/ +0.4041/ +0.7891$ against $+0.9296/ +0.9352/ +0.4982/ +0.4040/ +0.7891$ — which is also the third of three independent bit-level reproductions of that cell across separate extractions, the others being the sweep’s own control arm and the graded set’s. Nothing else in the sweep was affected: every other arm’s rendered spacing was verified against its construction before extraction.

Several diagnostic interventions were motivated by patterns observed during the study. We therefore report their provenance explicitly rather than treating confirmatory tests, precommitted follow-up analyses, and post-hoc diagnostics as equivalent forms of evidence.

For each diagnostic associated with the remote-evidence analyses, including the distance sweep whose reading was later withdrawn, the proposed control and the interpretation assigned to its possible outcomes were recorded before the corresponding run was executed.

Stage 5 is the case in which the proposed cross-edit diagnostic failed its own positive control. Stages 7 and 8 are the two source attributions this paper withdraws: drawn distance, which no arm separates on, and arrowhead exposure, which does not survive being measured on the rendered image. Stage 10 is the correlational attention measurement, whose ordering of the arms varies with depth. Stage 11 identifies a dependency on the implicated region through selective masking. It is what identifies the path carrying the signal that stage 5 had been introduced to test, and stage 12 is the audit of the baseline that stage 11’s table is read against.

The ordering matters for interpretation. The anomalous signal was observed before the diagnostic intended to explain it was proposed, and the positive control that invalidated the cross-edit criterion was introduced subsequently. We therefore distinguish these diagnostic follow-ups from the pre-registered confirmatory tests and report their provenance explicitly.

Together, the record documents not only the final controlled result but the sequence of interventions required to identify which sources of predictive information could and could not support the apparent graph-role signals.

K RENDERING COMPARISONS AT THREE BAND BLOCKS

Section 4.3 quotes the ten rendering arms at block 38 only. Table 13 condenses the same arms at three blocks of the Qwen-Image band, so that the separation between the arms that move the arrow and the arms that redraw it can be read at more than one depth. It is a condensation of Table 3 in Appendix D, which prints all five blocks, the confidence intervals and the per-arm construction details; nothing here is a separate measurement. Two features are worth reading off it directly. The four arms that change only where the arrow is drawn hold between $+0.458$ and $+0.980$ bits across all three blocks, including the arm drawn 153 pixels farther away, which is the strongest row at every block. The seven arms that change how it is drawn sit between $+0.006$ and $+0.095$ bits, and the ordering within that group runs *against* measured arrowhead exposure: at block 38 the fully covered head (exposure 0.00) reads $+0.071$ bits, higher than the shaft-stretched arm (0.93) at $+0.019$ and the tight-span redraw (0.95) at $+0.021$. That inversion is the observation behind the withdrawal of the exposure account in Appendix O.

stage	observation	control or intervention	outcome	record
1	apparent role signal	identical-icon subset	entity identity	post-hoc
2	larger MLP signal	role $\times$ chain-length crossing	chain length	post-hoc
3	narrow displaced interval	label shuffle	estimator bias	post-hoc
4	gate passes, role at chance	accuracy vs. majority rate	blind control	post-hoc
5	one deep reversal cell	cross-edit invariance	criterion withdrawn	post-hoc
6	five-block band	dense depths, both strengths	reversal- and s -specific	precommitted
7	band carried by $n = 4$	distance-controlled stretch	drawn distance, withdrawn at stage 8	precommitted
8	distance sweep flat to 416 px	visibility $\times$ distance	arrowhead exposure, withdrawn at stage 9	precommitted
9	tight-span arms not uniformly covered	exposure measured on pixels; ten arms	a class of re-renderings; no single property isolated	precommitted (Amendment 15)
10	enrichment tracks the band	enrichment on all ten arms, and on the edited images	magnitude does not order the arms	precommitted, per-arm predictions (Amendment 16)
11	three re-renderings each end the signal	key masking, head vs ctrl vs none	the routing is causally necessary, in two models	entry written after the run (Amendment 17); null-arm and SD3.5 cuts precommitted (Amendment 18)
12	empty-mask control 0.2 bits below the frozen cell	re-probe the frozen features on the same three layouts	layout count, not a perturbed forward pass	precommitted (Amendment 18a)
13	the sharpest magnitude cell rests on 8 items	re-capture the shaft-stretched arm at 56×3	block-35 peak withdrawn; block-38 inversion confirmed	<i>not</i> precommitted; run after the 8-item read
14	yield inferred from a mismatch between two arms	grade the cut, $f \in \{0, .25, .5, .75, 1\}$, two arms	access sensitivity measured within a condition	precommitted, five predictions (Amendment 20)

Table 12: **Provenance of the diagnostic sequence.** Each stage records the observation motivating the next control or intervention, the explanation supported by that analysis, and whether the analysis was committed before its numbers were read. Stages 7 and 8 are the two source attributions this paper withdraws, stage 13 withdraws a third and smaller reading, and stage 8’s control is named here as it was named when it ran. Stage 13 is the only entry in this table for which no outcome was committed before its numbers were read. Stage 14 was motivated by the stage 13 comparison but tests access quantity rather than attention magnitude. It does not convert stage 13’s exploratory ranking observation into a precommitted result.

L SUPPORTING ATTENTION COMPARISONS

We additionally compare raw attention enrichment with decodability across rendering arms. These measurements describe whether two observables order the tested conditions similarly. They are separate from the causal access result and do not test the correspondence between attention rank and intervention impact within an arm.

Before intervention, we measured the attention mass that candidate-region queries assign to the role-defining arrowhead (Section 3). Attention enrichment over a uniform baseline broadly separates the positive and null arms beginning around block 35. The ordering of individual conditions varies across depth.

Enrichment for all ten arms is reported in Table 9, on a common protocol of 8 base items $\times$ 1 layout. Eight items is too thin to rest this section on, so we re-ran the capture on the shaft-stretched arm at

Table 13: **Ten arms, three blocks of the Qwen-Image late-middle band**, role ΔS / accuracy, six layout permutations. Rows above the rule change only *where* the arrow is drawn; rows below change *how* it is drawn. Every construction-validity gate passes. * marks a cell whose label-shuffle interval excludes zero; in the three blocks shown the flags fall in the patched and crowded arms, with offsets of -0.003 to -0.009 bits against effects of $+0.07$ to $+0.09$, so they point the wrong way to manufacture these estimates. The unmodified baseline arm is also flagged, at block 41, which this condensation does not show. The causal claim rests on the masking conditions of Table 11 rather than on these cells. Blocks 41 and 44 and the full caption are in Table 3, Appendix D.

arm	gap / exposure	35	38	47
none	263 / 0.92	+0.930 / .987	+0.935 / .993	+0.789 / .949
moved farther	416 / 0.92	+0.967 / 1.00	+0.969 / 1.00	+0.980 / 1.00
patch past tip	263 / 0.92	+0.696 / .915	+0.934 / .995	+0.458* / .827
redrawn on top	263 / 0.95	+0.930 / .985	+0.950 / 1.00	+0.829 / .961
head 35% covered	263 / 0.65	+0.095* / .644	+0.094* / .590	+0.043* / .611
head 68% covered	263 / 0.32	+0.061* / .611	+0.081* / .608	+0.020* / .564
head fully covered	263 / 0.00	+0.045* / .586	+0.071* / .612	+0.013* / .539
shaft 91 $\rightarrow$ 221 px	264 / 0.93	+0.075* / .605	+0.019* / .547	+0.011* / .548
span tightened	238 / 0.79	+0.008 / .545	+0.016 / .559	+0.022 / .583
span tightened, far	443 / 0.79	+0.006 / .531	+0.008 / .548	+0.024 / .586
span tightened, top	238 / 0.95	+0.024* / .563	+0.021* / .568	+0.028* / .607

the 56 items $\times$ 3 layouts already used for the four distance-contrast arms (Table 10, 168 passes per arm). That scale-up was run after the eight-item result was seen and is not pre-registered; we mark it as exploratory in the provenance table and report what it changes as well as what it confirms.

It changes one claim. On eight items the shaft-stretched arm reads 2.50 at block 35, above every positive arm. At 168 passes the same cell reads 2.15 ± 0.22 , which is not separable from the 416-pixel arm’s 2.17 ± 0.34 and sits below the 263-pixel arm’s 2.45 ± 0.36 . We therefore withdraw the reading that this arm carries the strongest routing we measure.

The larger capture retains an inversion of the point estimates at block 38, our reference block for the source-identification comparisons. The shaft-stretched arm reaches enrichment 1.88 ± 0.16 while carrying $\Delta S = +0.019$ bits, against 1.63 ± 0.25 for the 416-pixel arm carrying $+0.969$ bits. Thus, the arm with more decodable role information has the lower mean enrichment in this comparison. The two remaining null arms sit at 1.60 and 1.64 in the same cell, so three arms lying within 0.011 bits of one another span 1.60 to 1.88 in enrichment. Spreads are standard deviations across the 168 passes, not confidence intervals for the between-arm difference. The shaft-stretched scale-up is exploratory.

The inversion is localized in depth. At blocks 41 and 47 enrichment orders the arms in the expected direction: the positive arms rise to 2.56 and 3.53 while the shaft-stretched arm falls to 1.49 and 1.55. The result therefore does not support a depth-general failure of attention magnitude. Likewise, the tight-span arm in which the role edge is redrawn above the icons, not the wide-span redraw that keeps its signal at $+0.950$ bits, carries only $+0.021$ bits despite substantial enrichment on the eight-item protocol. We retain this smaller observation as descriptive evidence only.

The across-model comparison is also descriptive. At matched relative depth, SD3.5 reaches attention enrichment 4.12 compared with 1.93 in Qwen-Image, approximately $2.1\times$ stronger routing, while its peak role estimate is only $+0.021$ bits compared with $+0.943$ bits in Qwen-Image. Because SD3.5 lacks a positive role arm, this comparison cannot distinguish a weak information source from an inefficient path. Taken together, these observations show that raw enrichment and role decodability need not order these rendering conditions alike at every depth. The selective and graded cuts establish the readout’s dependency independently of that descriptive mismatch.

M EXPLORATORY MASKING OBSERVATIONS

That arm carries the only masking results in the study that fall outside the pre-registered outcomes, and we flag them here rather than only in the appendix. First, the small elevation it shows at block 35,

+0.0460 bits with CI_{90} [+0.0294, +0.0632] and inside the equivalence bound, is removed to chance by the head cut, -0.0021 [-0.0087, +0.0043], with non-overlapping intervals; the null in that arm is therefore weaker at block 35 than a single peak-block quote would suggest. Second, the control cut *raises* the same cell to +0.0990 [+0.0832, +0.1147], landing on the equivalence bound. Both cut cells carry a label-shuffle flag, with offsets of -0.008 and -0.011 bits; because those offsets are negative they displace an estimate downward and so cannot produce the increase we are reporting. Our pre-registration named the control cut *collapsing* as the outcome that would invalidate the intervention; an increase was not among the branches. Softmax competition is the natural account, since masking keys renormalizes the remaining mass toward the role arrowheads, and it is untested, so we do not interpret the increase mechanistically. It strengthens rather than weakens the specificity claim: a key-count account predicts that both cuts degrade the estimate, and here two comparable cuts move it in opposite directions in the same arm, at the same block, through the same code path. A 0.099-bit effect at one block of eight, in an arm null everywhere else, is a lead worth one experiment rather than a result.

N ADDITIONAL CONTROL-BATTERY RESULTS

After the remote information path is identified, the remaining graph-role estimates can be interpreted in the context of the full control battery. Four additional measurements that initially appeared relevant to graph role admit simpler explanations: entity identity and chain length act as stimulus shortcuts, while a displaced estimator and cross-edit transfer criterion fail their own measurement controls. Each failure is documented together with its corresponding positive control in Appendix G.1.

Once these alternatives are controlled, recoverable graph-role information is negligible wherever the measurement is readable. Importantly, this is not a weak-instrument result. On the same rows and cross-validation folds, the readout recovers the construction-validity contrast at +0.915 bits for Qwen-Image and +0.908 bits for SD3.5, while the corresponding graph-role estimates are only $+3 \times 10^{-5}$ and $+3 \times 10^{-6}$ bits at accuracy 0.500. Thus, the activations contain information that the measurement can recover, but the controlled graph-role contrast contributes essentially no additional predictive information.

This pattern holds across eight sampled depths and both renoising strengths in Qwen-Image and at every SD3.5 depth for which the validity gate passes. SD3.5's deep blocks provide the only clean intervals excluding zero, reaching +0.021 bits, but this value remains one fifth of the pre-specified $\delta = 0.10$ equivalence bound and is the same small signal removed by the intervention in Section 4.4.

A second topological family points the same way, with a caveat we state rather than bury. In a pre-registered construction where one candidate edge feeds a directed cycle and the other feeds a bypass while node count, edge count, rendered ink, and the primary family's role label are held fixed, the role estimate ranges from -0.0011 to +0.0020 bits at chance accuracy across the same late blocks while the validity gate passes. That family is a weaker instrument than the primary one: it recovers the construction-validity contrast at +0.43 to +0.55 bits against roughly +0.9 bits here, so it is underpowered for the comparison it was built to make and the null it returns is a null under a less sensitive measurement. We report it as a generalization attempt that did not find a signal, not as evidence that no other role would. Additional decoder and pooling variants lead to the same overall conclusion: a regularized MLP classifies the validity contrast at accuracy 1.000 while leaving graph-role information at approximately $\pm 2 \times 10^{-5}$ bits, and the pairwise-endpoint and AI2D-RST analyses provide further corroboration where their own interpretation criteria permit it.

We therefore read the final result narrowly. After controlling position, appearance, endpoint identity, local geometry, global chain structure, and the identified remote information path, graph role is not usefully recoverable under the readouts, depths, timesteps, perturbations, and models evaluated here. The result does not establish architecture-independent absence of graph structure; it shows that the strong apparent positive can be explained by another information source, while the remaining controlled measurements provide negligible evidence for recoverable graph role.

O WITHDRAWN INTERPRETATIONS

Three explanations we published to ourselves along the way did not survive the next control. Section 4.6 states them and the principle they illustrate; this appendix gives each one in full rather than re-raising it where it arose. All three are recorded with their dates and their pre-registration status in Table 12.

Drawn distance. The first account of the band was that the readout has a spatial range and the role-defining arrow sits inside it. The distance sweep refutes it: the estimate does not decay from 263 to 416 pixels and the farthest arm is among the strongest (Section 4.3).

Arrowhead exposure. The second account was that the collapse tracks how much of the arrowhead is visible. Measuring exposure on the finished images refutes it: templates whose arrowhead is essentially unoccluded null as hard as the rest, and the shaft-stretched arm nulls at full exposure (Appendix D).

The shaft-stretched arm as the strongest routing. On eight items that arm read 2.50 enrichment at block 35, above every positive arm. At 56 items $\times$ 3 layouts the same cell reads 2.15 ± 0.22 and the claim is withdrawn (Section 4.6).

What replaced the first two is a class of remote rendering evidence with no single measured property isolated, which we state as a limitation rather than a finding. What replaced the third is the better-powered inversion at block 38. The pattern across all three is the one the paper argues for: each reading was consistent with the data available when it was made, and each fell to a measurement that varied the thing it named.